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Receiver circuit

Abstract

A receiver circuit comprising: an input-pin; a receiver-input-node; a ground-pin; an internal-node that is connected to the input-pin; and a MOSFET. The MOSFET has a conduction channel connected in series between the internal-node and the receiver-input-terminal; and a gate terminal, the voltage at which sets the conductivity of the conduction channel. The receiver circuit also includes an amplifier that: has an input terminal that is connected to the internal-node; and provides a voltage control signal to the gate terminal of the MOSFET such that the voltage at the internal-node with respect to the ground-pin is constant.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the priority under 35 U.S.C. § 119 of European Patent application no. 22306256.3, filed on 25 Aug. 2022, the contents of which are incorporated by reference herein.

FIELD

(2) The present disclosure relates to receiver circuits, and in particular to receiver circuits for processing infrared signals received at a photo-diode.

SUMMARY

(3) According to a first aspect of the present disclosure there is provided a receiver circuit comprising: an input-pin; a receiver-input-node; a ground-pin; an internal-node that is connected to the input-pin; a MOSFET having: a conduction channel connected in series between the internal-node and the receiver-input-node; and a gate terminal, the voltage at which sets the conductivity of the conduction channel; and an amplifier that: has an input terminal that is connected to the internal-node; and provides a voltage control signal to the gate terminal of the MOSFET such that the voltage at the internal-node with respect to the ground-pin is constant.

(4) Advantageously, the MOSFET and the amplifier can improve the performance of the receiver circuit with respect to ground noise injection.

(5) In one or more embodiments, the MOSFET is a PMOS MOSFET.

(6) In one or more embodiments, the MOSFET is an NMOS MOSFET.

(7) In one or more embodiments, the receiver circuit may further include a reference-voltage-source that is configured to provide a reference voltage to another input terminal of the amplifier.

(8) In one or more embodiments: the receiver circuit includes a first-ground-pin and a second-ground-pin; the receiver circuit is configured to process a signal at the receiver-input-node with

respect to a voltage on the first-ground-pin; and the reference-voltage-source is provided with respect to a voltage on the second-ground-pin.

(9) In one or more embodiments: the receiver circuit is configured to process a signal at the receiver-input-node with respect to a voltage on the ground-pin; and the reference-voltage-source is provided with respect to a voltage on the ground-pin.

(10) In one or more embodiments, the receiver circuit further comprises: a first-current-source connected between a supply-voltage-terminal and one end of the conduction channel of the MOSFET; and a second-current-source connected between a ground-pin and the other end of the conduction channel of the MOSFET.

(11) In one or more embodiments: the first-current-source is connected between the supply-voltage-terminal and the source of the MOSFET; and the second-current-source is connected between the ground-pin and the drain of the MOSFET.

(12) In one or more embodiments, the internal-node is galvanically connected to the input-pin by a bond wire.

(13) In one or more embodiments, the input-pin is configured to receive a current signal from a sensor.

(14) In one or more embodiments, the sensor comprises a photo-diode.

(15) According to a further aspect of the present disclosure, there is provided an integrated circuit comprising any receiver circuit disclosed herein.

(16) According to a further aspect of the present disclosure, there is provided a system comprising: any receiver circuit disclosed herein provided on an integrated circuit; and a sensor, which is external to the integrated circuit, wherein the sensor is configured to provide a current signal to the input-pin of the receiver circuit.

(17) In one or more embodiments, the sensor comprises a photo-diode.

(18) In one or more embodiments, the system is a battery management system.

(19) While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that other embodiments, beyond the particular embodiments described, are possible as well. All modifications, equivalents, and alternative embodiments falling within the spirit and scope of the appended claims are covered as well.

(20) The above discussion is not intended to represent every example embodiment or every implementation within the scope of the current or future Claim sets. The figures and Detailed Description that follow also exemplify various example embodiments. Various example embodiments may be more completely understood in consideration of the following Detailed Description in connection with the accompanying Drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) One or more embodiments will now be described by way of example only with reference to the accompanying drawings in which:
- (2) FIG. 1 shows an example of a receiver circuit;
- (3) FIG. 2 shows the receiver circuit of FIG. 1 that will be used to describe the effects of ground noise;
- (4) FIGS. 3a and 3b illustrate simulation results for the receiver of FIG. 1;
- (5) FIGS. 4a, 4b and 4c illustrate simulation results for the receiver of FIG. 2;
- (6) FIG. 5 illustrates an example embodiment of a receiver circuit 500 according to the present disclosure;
- (7) FIGS. 6a and 6b illustrate simulation results for the receiver circuit of FIG. 5, but without the

switching noise voltage source, $V_{\text{sub.SW}}$;

(8) FIGS. **7a**, **7b** and **7c** illustrate simulation results for the receiver circuit of FIG. **5** with the switching noise voltage source, $V_{\text{sub.SW}}$;

(9) FIG. **8** illustrates another example embodiment of a receiver circuit according to the present disclosure;

(10) FIG. **9** illustrates a yet further example embodiment of a receiver circuit according to the present disclosure;

(11) FIGS. **10a** and **10b** illustrate simulation results for the receiver circuit of FIG. **9**, but without the switching noise voltage source, $V_{\text{sub.SW}}$; and

(12) FIGS. **11a**, **11b** and **11c** illustrate simulation results for the receiver circuit of FIG. **9** with the switching noise voltage source, $V_{\text{sub.SW}}$.

DETAILED DESCRIPTION

(13) FIG. **1** shows an example of a receiver circuit **100**. In this example, the receiver circuit is provided on an integrated circuit (IC), a boundary of which is shown with a dashed line **101** in FIG. **1**. Two of the pins of the IC are shown in FIG. **1**: an input-pin, rx_in, **102** and a ground-pin, vssa, **103**.

(14) The receiver circuit **100** is for an infra red (IR) communication system, in this example an IR communication system for a battery management system (BMS). Such an IR communication system may be a serial IR system. A photo-diode **104** is shown connected to the input-pin, rx_in, **102**. The photo-diode **104** receives an IR communication signal from a corresponding IR transmitter. The photo-diode **104** is connected to a low power voltage source (VLP) **105**.

(15) Also shown in FIG. **1** is a capacitance, $C_{\text{sub.PCB}}$, **106** connected in series between the input-pin, rx_in, **102** and a printed circuit board (PCB) ground **107**. The ground-pin, vssa, **103** of the IC is also connected to the PCB ground **107**. This capacitance, $C_{\text{sub.PCB}}$, **106** represents the parasitic capacitance of the PCB to which the IC is connected.

(16) Returning now to the receiver circuit **100** on the IC, a receiver-input-node **108** is indicated in the figure. The inductance of a bond wire, $L_{\text{sub.bond}}$, **111** is shown connected between the input-pin, rx_in, **102** and the receiver-input-node **108**. An input-resistor **109**, $R_{\text{sub.IN}}$, and an input-capacitor, $C_{\text{sub.IN}}$, **110** are shown connected in parallel between the receiver-input-node **108** and the ground-pin, vssa, **103** (via another bond wire, $L_{\text{sub.bond}}$, **112**). The input-resistor **109**, $R_{\text{sub.IN}}$, and the input-capacitor, $C_{\text{sub.IN}}$, **110** are included to represent the input impedance of the receiver. Although not shown in FIG. **1**, it will be appreciated that the receiver circuit **100** includes additional components for providing a receiver-output-signal, rx_out.

(17) In practice, ground noise is generated inside the integrated circuit due to transient current consumption flowing through the bond wires, $L_{\text{sub.bond}}$, **111**, **112**. Current consumption can come from blocks sharing the same ground.

(18) FIG. **2** shows the receiver circuit of FIG. **1** that will be used to describe the effects of such ground noise. Features of FIG. **2** that have already been described with reference to FIG. **1** have been given corresponding reference numbers in the **200** series and will not necessarily be described again here.

(19) In FIG. **2**, a switching noise voltage source, $V_{\text{sub.SW}}$, **213** is shown to model a perturbation induced by other operating blocks (with conducted current or substrate noise injection (SNI)) sharing the same ground. The ground noise that is identified above, and that is modelled in FIG. **2** as the switching noise voltage source, $V_{\text{sub.SW}}$, **213**, can generate tens of mV of noise with respect to PCB ground **207** on the local IC grounds (ground-pin, vssa, **203** in FIG. **2**). Therefore, the voltage at the receiver-input-node **208** with respect to PCB ground **207** is modulated by this same ground noise level. In a system that is based on current detection (current mode communication) and where the receiver input stage is low impedance (such as the IR communication system of FIG. **2**), the ground noise can be “seen” by the receiver at the receiver-input-node **208** as an input signal depending on its amplitude and frequency. Therefore, this

unintended input current signal at the receiver-input-node **208** can generate an unintended (and undesirable) receiver output signal, rx_out, which can prevent the receiver from properly decoding the expected input signal that is provided by the photo-diode **204** (or any other external sensor, depending upon the communication system).

(20) One way to address this issue is for the receiver **200** to be grounded with a dedicated ground bonding. This would overcome any shared ground noise. However, this would be at a cost of an additional ESD (electrostatic discharge) protection and an additional package pin. This is because sharing the same package pin with several bondings will not prevent ground noise due to mutual bonding inductance.

(21) FIGS. **3a** and **3b** illustrate simulation results for the receiver of FIG. **1**; that is, one that doesn't have any noise.

(22) FIG. **3a** shows a simulation when there is no communication (that is, the photo-diode that is shown in FIG. **1** does not receive an IR communication signal). The upper plot of FIG. **3a** shows the current **314** at the input-pin, rx_in. The lower plot of FIG. **3a** shows the voltage **315** of the receiver-output-signal, rx_out. As expected, there is no noise present in either of the plots of FIG. **3a**.

(23) FIG. **3b** shows a simulation when there is high speed communication (that is, the photo-diode that is shown in FIG. **1** receives an IR communication signal). As with FIG. **3a**: the upper plot of FIG. **3b** shows the current **316** at the input-pin, rx_in; and the lower plot of FIG. **3b** shows the voltage **317** of the receiver-output-signal, rx_out. Again, as expected, there is no noise present in either of the plots of FIG. **3b**. The receiver-output-signal, rx_out, **317** provides a good representation of the input-signal that is provided by the photo-diode.

(24) FIGS. **4a**, **4b** and **4c** illustrate simulation results for the receiver of FIG. **2**; that is, one that has the switching noise modelled by a switching noise voltage source, V.sub.SW.

(25) FIG. **4a** shows a simulation of the voltage provided by the switching noise voltage source, V.sub.SW, **418** which in this example has a value of 36 mVpp (millivolts, peak to peak amplitude). As can be seen from FIG. **4a**, the noise is injected after 1 μ s.

(26) FIG. **4b** shows a simulation when there is no communication (that is, the photo-diode that is shown in FIG. **2** does not receive an IR communication signal). The upper plot of FIG. **4b** shows the current **419** at the input-pin, rx_in. The lower plot of FIG. **4b** shows the voltage **420** of the receiver-output-signal, rx_out. As can be seen from FIG. **4b**, the noise injected after 1 μ s is processed by the receiver such that it translates into a receiver-output-signal, rx_out, **420** that incorrectly indicates that communication data has been received by the photo-diode. That is, when Rx_in is perturbed by switching noise when there is no communication, the receiver is affected such that it outputs the wrong message.

(27) FIG. **4c** shows a simulation when there is high speed communication (that is, the photo-diode that is shown in FIG. **2** receives an IR communication signal). As with FIG. **4b**: the upper plot of FIG. **4c** shows the current **421** at the input-pin, rx_in; and the lower plot of FIG. **4c** shows the voltage **422** of the receiver-output-signal, rx_out. As can be seen from FIG. **4c**, the noise injected after 1 μ s degrades the quality of the receiver-output-signal, rx_out, **422**. That is, when Rx_in is perturbed by switching noise when there is communication, the receiver is also affected such that it outputs the wrong message.

(28) Examples of the present disclosure, that will be described below, can significantly improve high-speed current mode communication and the impact of receiver ground noise on the sensitivity of the receiver. Such examples can improve immunity from this noise, which is equivalent to an input current signal in current mode communication, especially where the receiver input stage is low impedance. Due to this noise immunity improvement, a receiver detection threshold can then be set to a lower current level and therefore can allow communication over longer distances between an emitter and the receiver (e.g. infrared communication with photo-diodes).

(29) FIG. **5** illustrates an example embodiment of a receiver circuit **500** according to the present

disclosure. Features of FIG. 5 that are also illustrated in an earlier drawing are given corresponding reference numbers in the **500** series and will not necessarily be described again here.

(30) The receiver circuit **500** has an input-pin, rx_in, **502** and a receiver-input-node, Rx_in3, **508** that are the same way as those described with reference to FIGS. **1** and **2**.

(31) In the example of FIG. **5** the IC that includes the receiver circuit **500** has two ground pins: a first-ground-pin, VSSC, **503** and a second-ground-pin, VSSA, **531**. The first-ground-pin, VSSC, **503** is a reference pin for communication blocks in the IC, including the receiver circuit **500**. The second-ground-pin, VSSA, **531** is a reference pin for other blocks in the IC. Providing two separate ground-pins **503**, **531** in this way can beneficially reduce the ground noise that is experienced at the ground-pin (VSSC, **503**) for the receiver circuit **500**. Nonetheless, as will be discussed below with reference to FIG. **9**, in other examples the IC may have only one ground-pin.

(32) In this example, the receiver circuit **500** also includes an internal-node, Rx_in2, **524** that is connected to the input-pin, rx_in, **502**. In FIG. **5** the internal-node, Rx_in2, **524** is galvanically connected to the input-pin, rx_in, **502**. However, in other examples the package may not require bond wires and therefore the internal-node, Rx_in2, **524** does not necessarily need to be galvanically connected to the input-pin, rx_in, **502**.

(33) As will be discussed below, the receiver circuit includes additional components (when compared with FIG. **1**), that are shown in box **523** in FIG. **5**, such that the internal-node, Rx_in2, **524** is a low impedance node that is connected to the sensor output (that is, the input-pin, rx_in, **502** of the IC that receives the output signal from the sensor—in this example the photo-diode **504**). As will be explained below, and as illustrated by way of simulation results, the internal-node, Rx_in2, **524** is not modulated by the switching noise voltage source, V.sub.SW, **513**.

(34) The additional components **523** include a metal-oxide-semiconductor field-effect transistor (MOSFET). In FIG. **5** a PMOS MOSFET **525** is shown. FIG. **8** illustrates how an NMOS MOSFET can be used instead.

(35) Returning to FIG. **5**, the PMOS MOSFET **525** has a conduction channel (between its source and its drain, as it known in the art) that is connected in series between the internal-node, Rx_in2, **524** and the receiver-input-node, Rx_in3, **508**. The PMOS MOSFET **525** also has a gate terminal, the voltage at which sets the conductivity of the conduction channel.

(36) The additional components **523** in this example also include an amplifier **526**. The amplifier **526** has an input terminal that is connected to the internal-node, Rx_in2, **524**. The amplifier **526** has an output terminal that provides a voltage control signal to the gate terminal of the PMOS MOSFET **525** such that the voltage at the internal-node, Rx_in2, **524** with respect to the first-ground-pin, VSSC, **503** is constant. In this way, any “ground” switching noise voltage will not cause a significant current to be received at the input-pin, rx_in, **502** from the parasitic capacitance, C.sub.PCB, **506**.

(37) In the example of FIG. **5**, the additional components **523** also include: a reference-voltage-source **527**, a first-current-source, I0, **529** and a second-current-source, I0, **530**.

(38) The reference-voltage-source **527** provides a voltage of Vref to another input terminal of the amplifier **526**. As shown in FIG. **5**, in this example, the reference-voltage-source **527** is provided with respect to the voltage on the second-ground-pin, VSSA, **531**. Whereas, the receiver circuit **500** is configured to process a signal at the receiver-input-node **508** with respect to a voltage on the first-ground-pin, VSSC, **503**. The value of Vref can be V_{sir}, where V_{sir} is the voltage provided by a serial infrared power supply. V_{sir} can be 1.5V with respect to vssa, in some examples. The specific value of Vref does not affect the functionality of the receiver circuit **500**, as long as Vref is larger than rx_in3 voltage (receiver input voltage). This is important for guaranteeing that the PMOS MOSFET **525** (pass device) has more than 100-200 mV drop out voltage (VDS).

(39) The first-current-source, I0, **529** is connected between a supply-voltage-terminal, Vsupply, and the internal-node, Rx_in2, **524**. Therefore, the first-current-source, I0, **529** provides a current to one end of the conduction channel of the PMOS MOSFET **525** (the source of the PMOS MOSFET **525**

in FIG. 5). The second-current-source, **I0, 530** is connected between a ground-pin (in this example the second-ground-pin, **VSSA, 531**) and the receiver-input-node, **Rx_in3, 508**. Therefore, the second-current-source, **I0, 530** provides a current to the other end of the conduction channel of the PMOS MOSFET **525** (the drain of the PMOS MOSFET **525** in FIG. 5). The first-current-source, **I0, 529** and the second-current-source, **I0, 530** are provided to ensure that the PMOS MOSFET **525** is correctly biased, even when the photo-diode **504** does not receive any IR light and therefore does not provide a sensed current signal, such that the PMOS MOSFET **525** is always active and can be appropriately operated such that the current through its conduction channel can be controlled.

(40) The additional components **523** create a new low impedance node (the internal-node, **Rx_in2, 524**) that is connected to the sensor output (via the input-pin, **rx_in, 502**) and referenced to a noise free ground (the second-ground-pin, **VSSA, 531**) while the receiver ground is connected to the communication ground (the first-ground-pin, **VSSC, 503**) which can be noisy due to current consumption of other communication blocks that are also connected to the same ground. As will be shown in the following simulation results, the voltage at the input-pin, **rx_in, 502** with respect to **pcb_ground 507** is still noisy but it does not generate a current input signal at the receiver output because the drain impedance of the PMOS MOSFET **525** is high enough to avoid or reduce current coupling.

(41) FIGS. **6a** and **6b** illustrate simulation results for the receiver circuit of FIG. 5, but without the switching noise voltage source, **V.sub.SW, 736**. FIGS. **7a, 7b** and **7c** illustrate simulation results for the receiver circuit of FIG. 5 with the switching noise voltage source, **V.sub.SW, 736**.

(42) FIG. **6a** shows a simulation when there is no communication (that is, the photo-diode that is shown in FIG. 5 does not receive an IR communication signal). The upper plot of FIG. **6a** shows the current **632** at the input-pin, **rx_in**. The lower plot of FIG. **6a** shows the voltage **633** of the receiver-output-signal, **rx_out**. As expected, there is no noise present in either of the plots of FIG. **6a**. Also, by comparing the lower plot of FIG. **6a** with the corresponding lower plot of FIG. **3a**, it can be seen that the inclusion of the additional components of FIG. 5 does not affect the voltage **633** of the receiver-output-signal, **rx_out**.

(43) FIG. **6b** shows a simulation when there is high speed communication (that is, the photo-diode that is shown in FIG. 5 receives an IR communication signal). As with FIG. **6a**: the upper plot of FIG. **6b** shows the current **634** at the input-pin, **rx_in**; and the lower plot of FIG. **6b** shows the voltage **635** of the receiver-output-signal, **rx_out**. By comparing the plots of FIG. **6b** with the corresponding plots of FIG. **3b**, it can be seen that the inclusion of the additional components of FIG. 5 cause the current **634** at the input-pin, **rx_in**, to have a slightly different shape but that the receiver-output-signal, **rx_out, 635** is unaffected. That is, the inclusion of the additional components in FIG. does not degrade the performance of the receiver circuit when there is no noise present.

(44) Turning now to FIGS. **7a, 7b** and **7c**, FIG. **7a** shows a simulation of the voltage provided by the switching noise voltage source, **V.sub.SW, 736**. In the same way as FIG. **4a**, the noise in FIG. **7a** is injected after 1 μ s (purely as an example).

(45) FIG. **7b** shows a simulation when there is no communication (that is, the photo-diode that is shown in FIG. 5 does not receive an IR communication signal). The upper plot of FIG. **7b** shows the current **737** at the input-pin, **rx_in**, which includes the noise that is injected by the switching noise voltage source, **V.sub.SW, 736**. The lower plot of FIG. **7b** shows the voltage **738** of the receiver-output-signal, **rx_out**. As can be seen from FIG. **7b**, the noise injected is not present in the receiver-output-signal, **rx_out, 738**. That is, despite the **rx_in** perturbation due to switching noise when there is no communication, there is no impact on **rx_out**. Therefore, advantageously the additional components of FIG. 5 have improved the performance of the receiver circuit with respect to ground noise injection.

(46) FIG. **7c** shows a simulation when there is high speed communication (that is, the photo-diode that is shown in FIG. 5 receives an IR communication signal). The upper plot of FIG. **7c** shows the

current **739** at the input-pin, rx_in; the middle plot of FIG. 7c shows the voltage **740** of the receiver-output-signal, rx_out; and the lower plot of FIG. 7c shows the voltage at the node in the circuit where the noise is injected. The lower plot in FIG. 7c is included to show that there is indeed noise in the system, even though it is not visible in the receiver-output-signal, rx_out, **740**. (47) As can be seen from FIG. 7c, the noise injected after 1 μ s does not degrade the quality of the signal at the input-pin, rx_in, **739** or the receiver-output-signal, rx_out, **740**. Since no switching noise is visible on rx_in current, the receiver circuit can correctly generate the message communication (rx_out).

(48) FIG. 8 illustrates another example embodiment of a receiver circuit **800** according to the present disclosure. The receiver circuit of FIG. 8 is very similar to the one of FIG. 5.

(49) In FIG. 8, an NMOS MOSFET **825** is used instead of the PMOS MOSFET of FIG. 5. Also, the first-current-source, **I0, 829** and the second-current-source, **I0, 830** are reversed in order to provide the correct biasing for the NMOS MOSFET **825**. Therefore, the first-current-source, **I0, 829** is connected between a supply-voltage-terminal, Vsupply, and the receiver-input-node, Rx_in3, **808**. In this way, the first-current-source, **I0, 829** provides a current to one end of the conduction channel of the NMOS MOSFET **825** (the drain of the NMOS MOSFET **825** in FIG. 8). The second-current-source, **I0, 830** is connected between a ground-pin (in this example the second-ground-pin, VSSA, **831** and the internal-node, Rx_in2, **824**. Therefore, the second-current-source, **I0, 830** provides a current to the other end of the conduction channel of the NMOS MOSFET **825** (the source of the NMOS MOSFET **825** in FIG. 8).

(50) FIG. 9 illustrates a yet further example embodiment of a receiver circuit **900** according to the present disclosure. The receiver circuit of FIG. 9 is similar to the one of FIG. 5. Features of FIG. 9 that have already been described with reference to FIG. 5 will not necessarily be described again here.

(51) In FIG. 9 the reference-voltage-source **927** is provided with respect to the voltage on the first-ground-pin, VSSC, **903**. That is, the first-ground-pin, VSSC, **903** is the reference-pin for the reference-voltage-source **927** and also for the receiver circuit **900**. In this way, the receiver circuit is configured to process a signal at the receiver-input-node **908** with respect to a voltage on the first-ground-pin, VSSC, **903**.

(52) The additional components **923** of FIG. 9 create the new low impedance node (the internal-node, Rx_in2, **924**) that is connected to the sensor output and in this example is referenced to VSSC ground (the first-ground-pin **903**). In some ICs, multiple ground pins do not exist and therefore it is beneficial that the additional components **923** of this example can function adequately using only one ground pin (the first-ground-pin **903**) for all IC functions.

(53) It will be appreciated that an NMOS MOSFET can be used instead of the PMOS MOSFET **925** that is shown in FIG. 9. This is in the same way that the NMOS MOSFET **825** of FIG. 8 can be used instead of the PMOS MOSFET **525** of FIG. 5.

(54) FIGS. 10a and 10b illustrate simulation results for the receiver circuit of FIG. 9, but without the switching noise voltage source, V.sub.SW. FIGS. 10a and 10b are very similar to FIGS. 6a and 6b, and will not be described in any detail here.

(55) FIGS. 11a, 11b and 11c illustrate simulation results for the receiver circuit of FIG. 9 with the switching noise voltage source, V.sub.SW. FIGS. 11a, 11b and 11c are very similar to FIGS. 7a, 7b and 7c.

(56) It can be seen from FIG. 11b that the switching noise injected after 1 μ s, as shown by the perturbation in rx_in **1142**, does not have a significant impact on rx_out **1143**.

(57) It can be seen from FIG. 11c that the noise injected after 1 μ s only very slightly degrades the quality of signal the input-pin, rx_in, **1144**, which in turn slightly degrades the quality of the receiver-output-signal, rx_out, **1145**. Nonetheless, the receiver-output-signal, rx_out, **1145** still represents good communication despite the small switching noise that was impacting the communication without the additional components of FIG. 5.

(58) Examples disclosed herein include circuitry that is added in series between a current mode receiver input pin and receiver input stage (which can be a trans-diode connected MOS for current communication, but not limited to this) that allows the receiver to operate with noise on its ground (without generating an unintended signal on its output). This circuitry can artificially create a new low impedance node connected to the sensor output (Integrated Circuit: IC input pin) that is not modulated by IC internal ground noise used to ground the receiver. Therefore, the receiver input stage current is much less modulated by its ground noise, resulting in the processing of only the current signal generated by the external sensor (e.g. photo-diode). It can enable communication over longer distances between the emitter and receiver because the receiver input current detection threshold can be set to a lower level without the risk of detecting an unwanted noise signal.

(59) The receivers of embodiments of the present disclosure can be current mode receivers, rather than voltage mode (that can include a high impedance differential amplifier). Therefore, applications that use a photo-diode as a sensor can be particularly well-suited to embodiments of the present disclosure because the useful information that is provided by the photo-diode is provided as a current. Such receivers of the present disclosure can provide good ground noise isolation and can also occupy relatively small areas on the die.

(60) The instructions and/or flowchart steps in the above figures can be executed in any order, unless a specific order is explicitly stated. Also, those skilled in the art will recognize that while one example set of instructions/method has been discussed, the material in this specification can be combined in a variety of ways to yield other examples as well, and are to be understood within a context provided by this detailed description.

(61) In some example embodiments the set of instructions/method steps described above are implemented as functional and software instructions embodied as a set of executable instructions which are effected on a computer or machine which is programmed with and controlled by said executable instructions. Such instructions are loaded for execution on a processor (such as one or more CPUs). The term processor includes microprocessors, microcontrollers, processor modules or subsystems (including one or more microprocessors or microcontrollers), or other control or computing devices. A processor can refer to a single component or to plural components.

(62) In other examples, the set of instructions/methods illustrated herein and data and instructions associated therewith are stored in respective storage devices, which are implemented as one or more non-transient machine or computer-readable or computer-usable storage media or mediums. Such computer-readable or computer usable storage medium or media is (are) considered to be part of an article (or article of manufacture). An article or article of manufacture can refer to any manufactured single component or multiple components. The non-transient machine or computer usable media or mediums as defined herein excludes signals, but such media or mediums may be capable of receiving and processing information from signals and/or other transient mediums.

(63) Example embodiments of the material discussed in this specification can be implemented in whole or in part through network, computer, or data based devices and/or services. These may include cloud, internet, intranet, mobile, desktop, processor, look-up table, microcontroller, consumer equipment, infrastructure, or other enabling devices and services. As may be used herein and in the claims, the following non-exclusive definitions are provided.

(64) In one example, one or more instructions or steps discussed herein are automated. The terms automated or automatically (and like variations thereof) mean controlled operation of an apparatus, system, and/or process using computers and/or mechanical/electrical devices without the necessity of human intervention, observation, effort and/or decision.

(65) It will be appreciated that any components said to be coupled may be coupled or connected either directly or indirectly. In the case of indirect coupling, additional components may be located between the two components that are said to be coupled.

(66) In this specification, example embodiments have been presented in terms of a selected set of details. However, a person of ordinary skill in the art would understand that many other example

embodiments may be practiced which include a different selected set of these details. It is intended that the following claims cover all possible example embodiments.

Claims

1. A receiver circuit comprising: an input-pin; a receiver-input-node; a ground-pin; an internal-node that is connected to the input-pin; a MOSFET having: a conduction channel connected in series between the internal-node and the receiver-input-node; and a gate terminal, the voltage at which sets the conductivity of the conduction channel; and an amplifier that: has an input terminal that is connected to the internal-node; and provides a voltage control signal to the gate terminal of the MOSFET such that the voltage at the internal-node with respect to the ground-pin is constant; a first-current-source connected between a supply-voltage-terminal and one end of the conduction channel of the MOSFET; and a second-current-source connected between a receiver circuit ground-pin and the other end of the conduction channel of the MOSFET.
2. The receiver circuit of claim 1, wherein the MOSFET is a PMOS MOSFET.
3. The receiver circuit of claim 1, wherein the MOSFET is an NMOS MOSFET.
4. The receiver circuit of claim 1, further including a reference-voltage-source that is configured to provide a reference voltage to another input terminal of the amplifier.
5. The receiver circuit of claim 4, wherein: the receiver circuit includes a first-ground-pin and a second-ground-pin; the receiver circuit is configured to process a signal at the receiver-input-node with respect to a voltage on the first-ground-pin; and the reference-voltage-source is provided with respect to a voltage on the second-ground-pin.
6. The receiver circuit of claim 4, wherein: the receiver circuit is configured to process a signal at the receiver-input-node with respect to a voltage on the ground-pin; and the reference-voltage-source is provided with respect to a voltage on the ground-pin.
7. The receiver circuit of claim 1, wherein the receiver circuit ground pin is the ground pin.
8. The receiver circuit of claim 1, wherein: the first-current-source is connected between the supply-voltage-terminal and the source of the MOSFET; and the second-current-source is connected between the receiver circuit ground-pin and the drain of the MOSFET.
9. The receiver circuit of claim 1, wherein the internal-node is galvanically connected to the input-pin by a bond wire.
10. The receiver circuit of claim 1, wherein the input-pin is configured to receive a current signal from a sensor.
11. The receiver circuit of claim 10, wherein the sensor comprises a photo-diode.
12. An integrated circuit comprising the receiver circuit of claim 1.
13. A system comprising: the receiver circuit of claim 1 provided on an integrated circuit; and a sensor, which is external to the integrated circuit, wherein the sensor is configured to provide a current signal to the input-pin of the receiver circuit.
14. The system of claim 13, wherein the sensor comprises a photo-diode.
15. The system of claim 13, wherein the system is a battery management system.
16. The receiver circuit of claim 5, wherein the receiver circuit ground pin is the second-ground-pin.
17. The receiver circuit of claim 4, wherein the internal-node is galvanically connected to the input-pin by a bond wire.
18. The receiver circuit of claim 4, wherein the input-pin is configured to receive a current signal from a sensor.
19. A system comprising: the receiver circuit of claim 4 provided on an integrated circuit; and a sensor, which is external to the integrated circuit, wherein the sensor is configured to provide a current signal to the input-pin of the receiver circuit.
20. A receiver circuit comprising: an input-pin; a receiver-input-node; an internal-node that is

connected to the input-pin; a MOSFET having: a conduction channel connected in series between the internal-node and the receiver-input-node; and a gate terminal, the voltage at which sets the conductivity of the conduction channel; and an amplifier that: has an input terminal that is connected to the internal-node; and provides a voltage control signal to the gate terminal of the MOSFET; a reference-voltage-source that is configured to provide a reference voltage to another input terminal of the amplifier; a first-current-source connected between a supply-voltage-terminal and one end of the conduction channel of the MOSFET; and a second-current-source connected between a receiver circuit ground-pin and the other end of the conduction channel of the MOSFET.
